EART11300

Physical properties of minerals

Video 3: Cleavage

Reading: Section on cleavage in the online mineralogy textbook:
https://opengeology.org/Mineralogy/3-properties/#352_Fracture_Cleavage_and_Parting

What is ‘cleavage’?

- Many minerals tend to break smoothly along particular planes. This is related to the internal atomic structure: it is a manifestation of weaker bond strength across these planes.
- Cleavage planes are always parallel to possible (or actual) crystal faces. Like crystal faces, they are related to the crystal’s internal symmetry.

Cleavage

- Cleavage is more useful as a diagnostic property than crystal shape, since it is not affected by the external conditions which determine crystal shape (the only limitation here being the size of the grains).
- It is an extremely valuable diagnostic property in thin sections of minerals viewed in the optical microscope.
- Cleavage may be developed in 1, 2, 3, 4 or 6 directions.
- Cleavage may be classified qualitatively as perfect, good, moderate or poor, and also in terms of the angular relationship between cleavage planes.
- Some minerals do not develop any cleavage.

Table showing the numbers and angles of cleavages possible in crystals

Number of cleavage directions	Description	Example on figure
0	No cleavage Mineral breaks into irregular masses with no flat surfaces	
1	Basal cleavage Mineral splits apart along flat sheets	A
2 at 90°	Prismatic cleavage Elongate with rectangular cross sections	B
2 not at 90°	Prismatic cleavage Elongate with parallelogram cross sections	C
3 at 90°	Cubic cleavage Mineral breaks into cubes	D
3 not at 90°	Rhombohedral cleavage Mineral breaks into rhombohedra	E
4	Octahedral cleavage Mineral breaks into octahedra	F
6	Dodecahedral cleavage Mineral breaks into dodecahedra	G

Note that the above terms refer to equivalent cleavage planes.

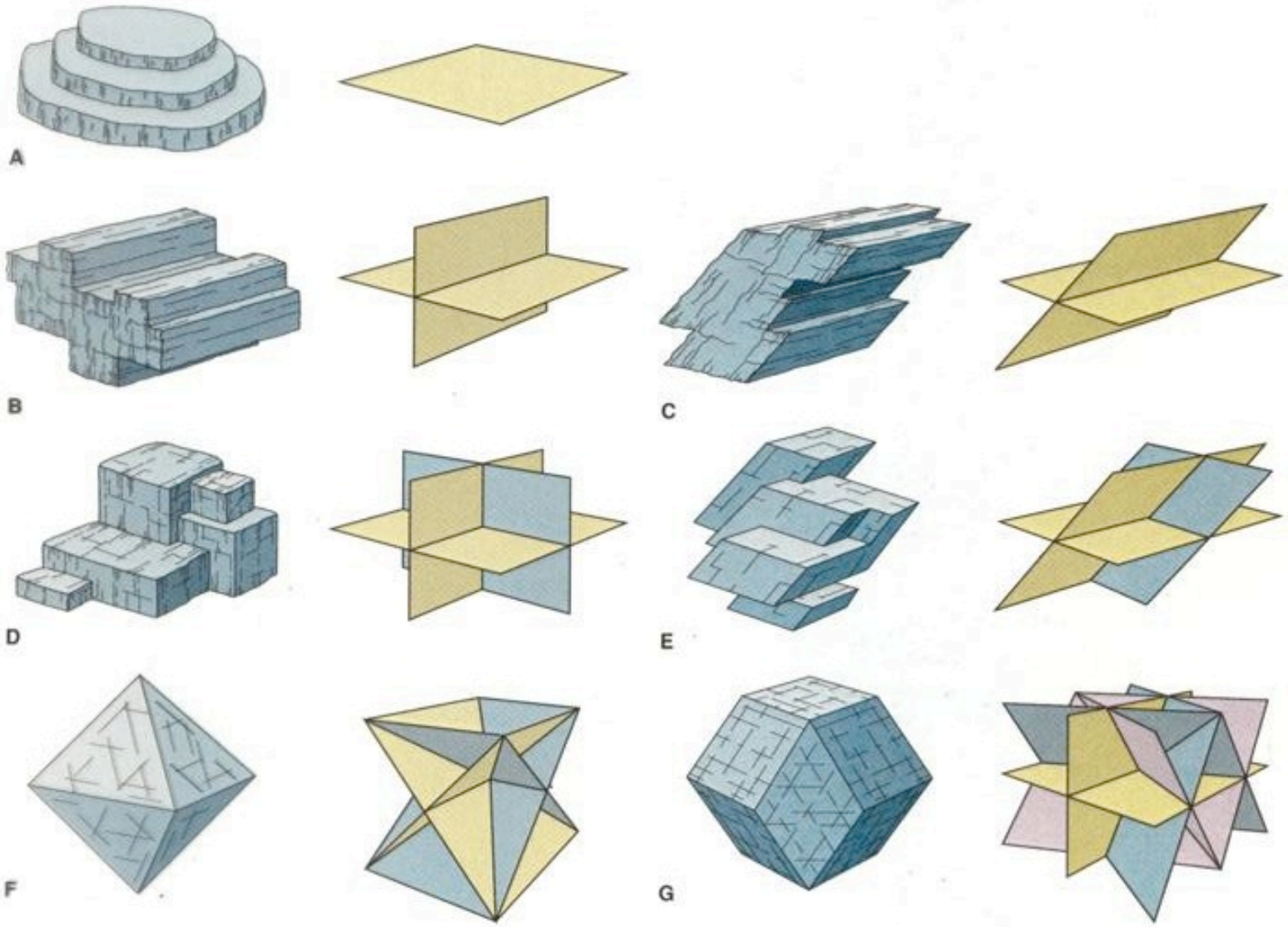
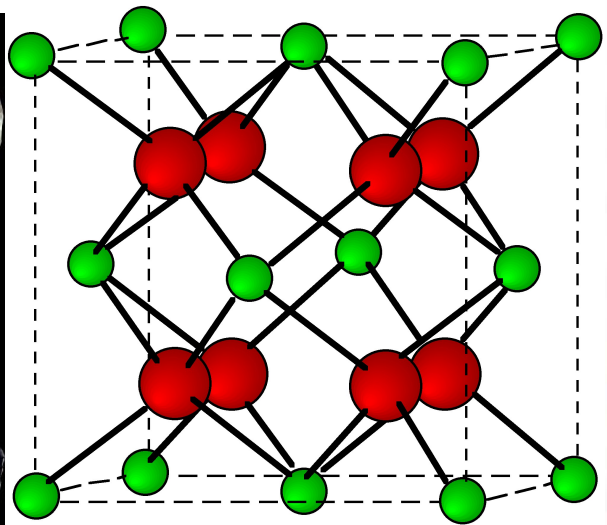


Figure from http://web.arc.losrios.edu/~borougt/MineralogyDiagrams.htm

Examples: 1. Fluorite CaF_2

- The photo on the left below shows cubic fluorite crystals.
- The photo on the right is of fluorite cleavage fragments. You can see that it has four equivalent perfect cleavages parallel to the faces of an octahedron.
- Which example on the figure does this correspond to?

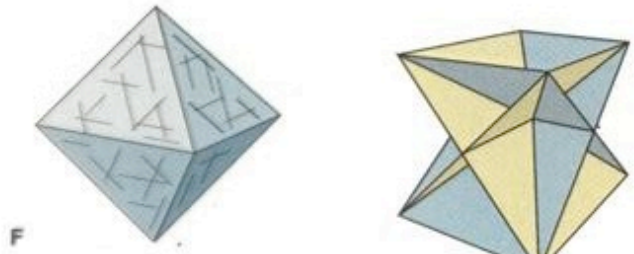


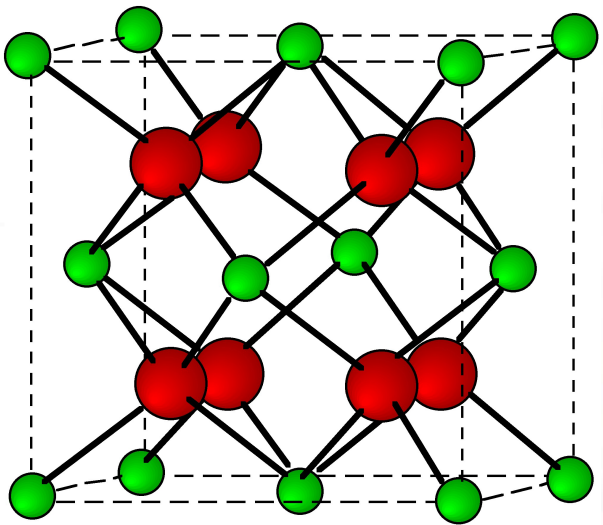




Examples: 1. Fluorite CaF_2

- The photo on the left below shows cubic fluorite crystals.
- The photo on the right is of fluorite cleavage fragments. You can see that it has four equivalent perfect cleavages parallel to the faces of an octahedron.
- The reason for the octahedral cleavage is that fluorite crystals break along planes where the bonding is weakest, which is equivalent to cutting off the corners on the model below (fluorite composition = CaF_2, green = Ca, red = F).

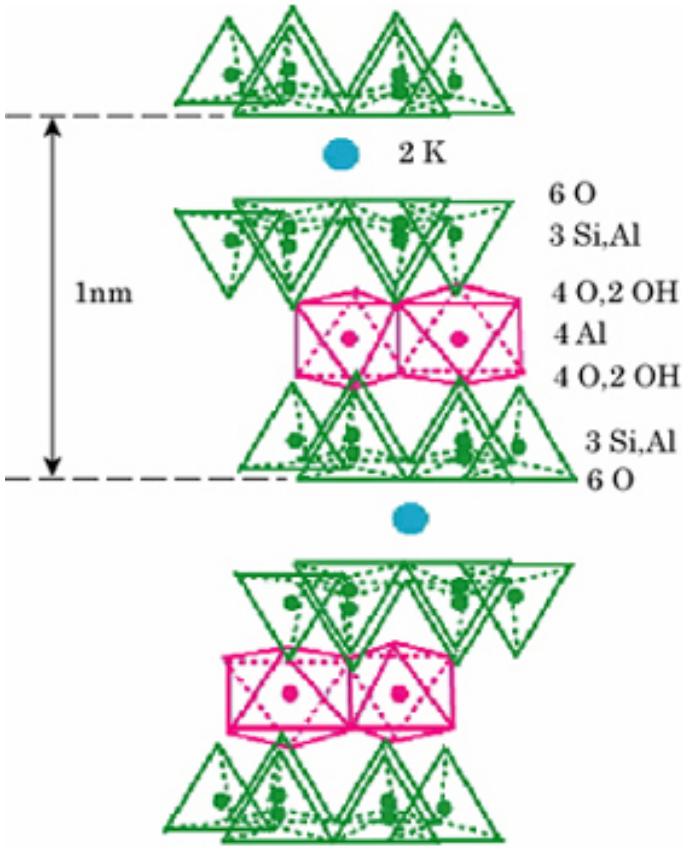






Examples: 2. Mica

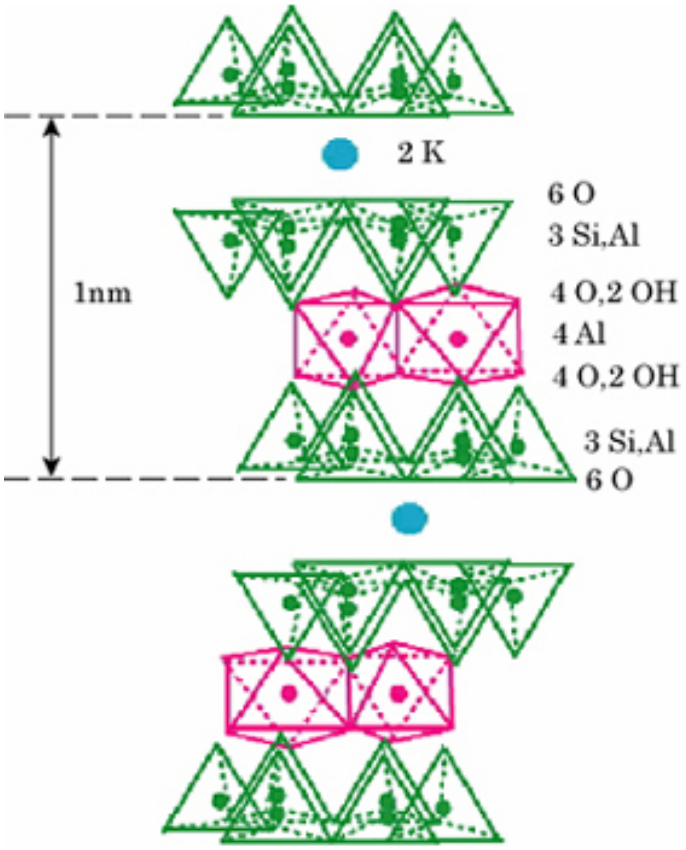
- **Mica** has one perfect cleavage.
- Which example on the figure does this correspond to?





Examples: 2. Mica

- **Mica** has one perfect cleavage.
- The reason for the perfect cleavage can be seen from the model of the structure below. The K^+ ions are weakly bonded to neighbouring cations, so the cleavage occurs in the layers containing the K^+ ions.



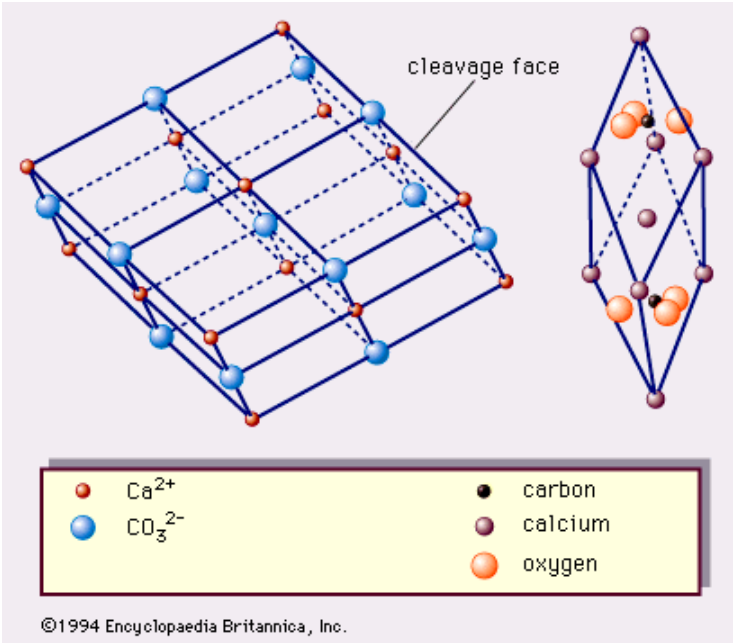




Examples: 3. Calcite, $CaCO_3$

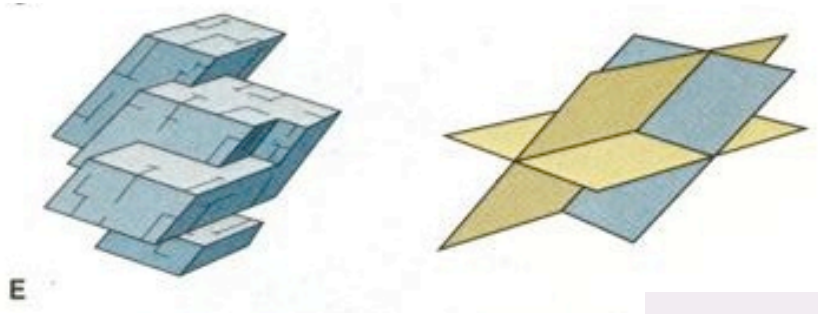
- **Calcite** has three equivalent perfect cleavages parallel to the faces of a rhombohedron.
- Which example on the figure does this correspond to?



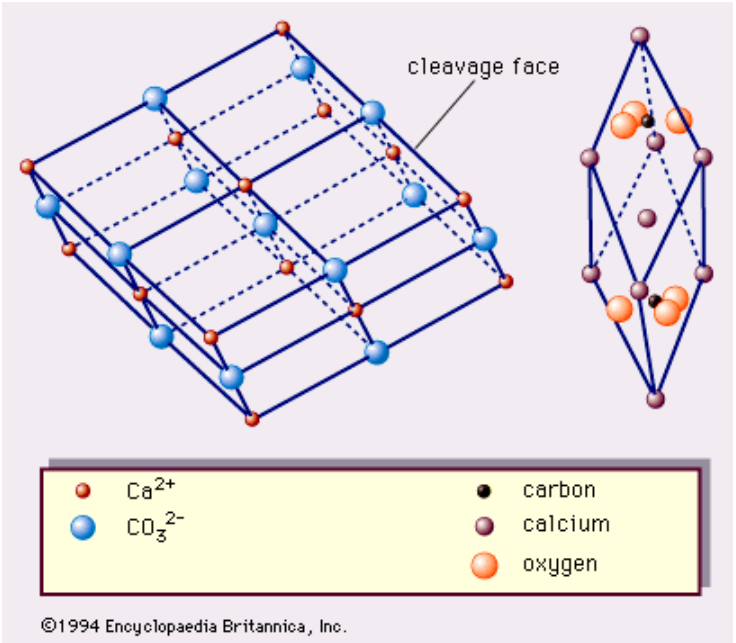


Examples: 3. Calcite, $CaCO_3$

- **Calcite** has three equivalent perfect cleavages parallel to the faces of a rhombohedron.
- The cleavage rhombohedron is different from the rhombohedral faces that are sometimes developed.



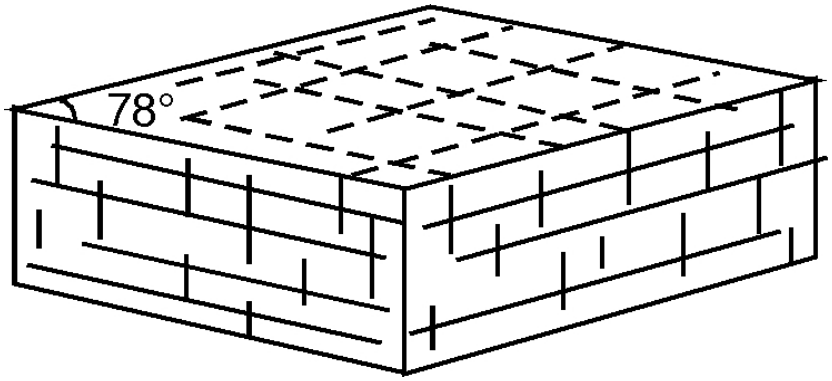


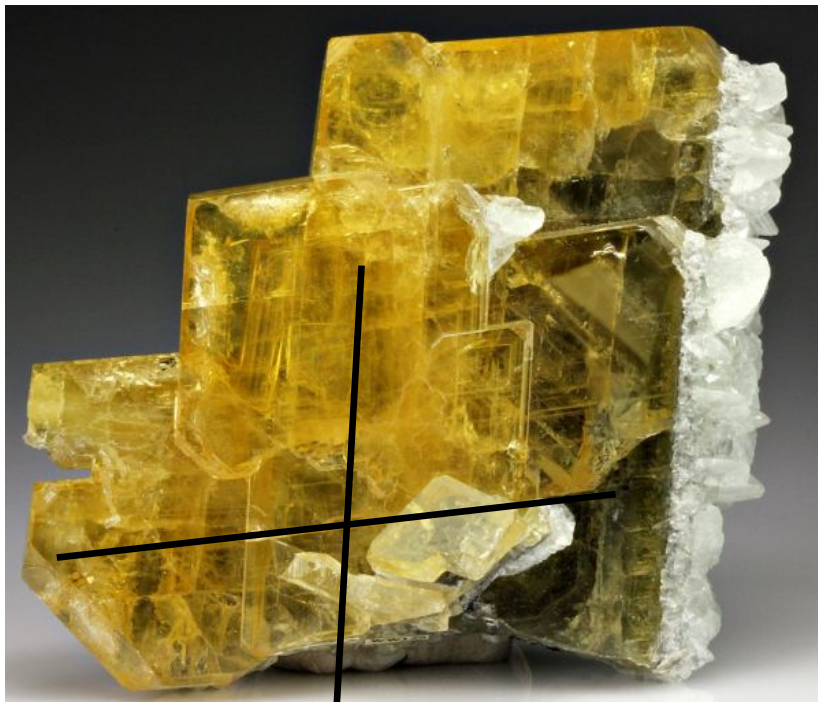


Examples: 4. Barite, $BaSO_4$

- **Barite** has two non-equivalent sets of cleavages. One set consists of one perfect cleavage. The other set contains two very good cleavages at 78° to each other and is perpendicular to the first set (see drawing).
- Which examples* on the figure does this correspond to?

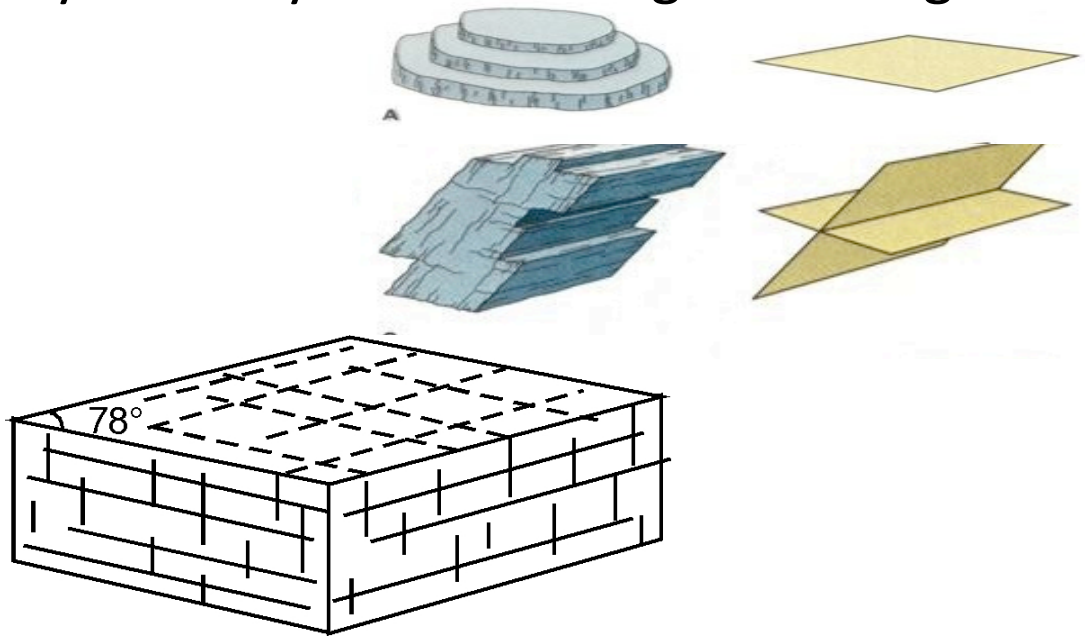
*It belongs to 2 different categories, as there are 2 non-equivalent sets of cleavages (each diagram in the figure shows a set of equivalent cleavages, i.e. ones that are related to each other by symmetry).

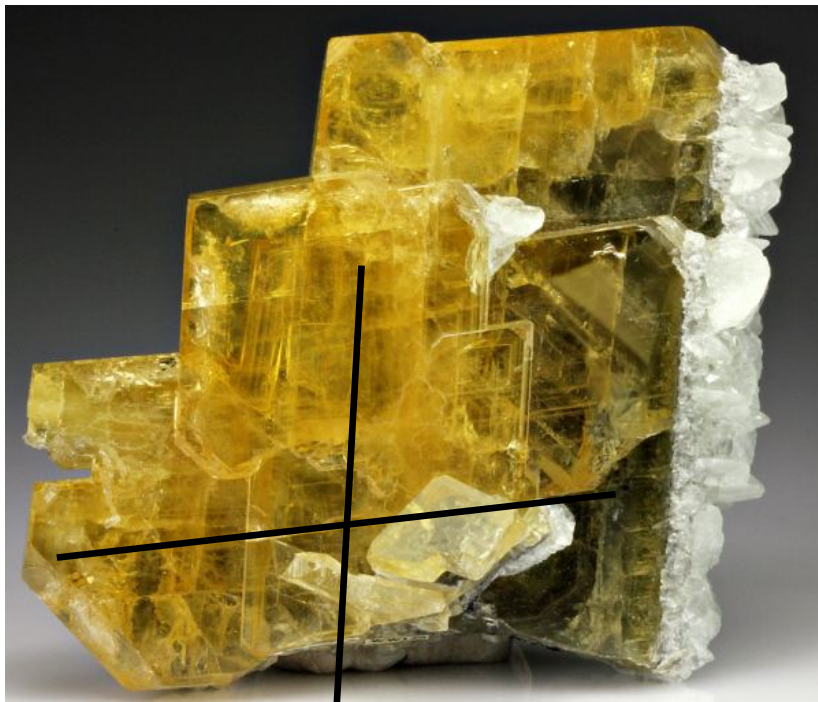




Examples: 4. Barite, $BaSO_4$

- **Barite** has two non-equivalent sets of cleavages. One set consists of one perfect cleavage. The other set contains two very good cleavages at 78° to each other and is perpendicular to the first set (see drawing).
- Notice how the cleavage lines on the drawing are shown to indicate that the cleavage occurs as cracks inside the crystal. You can also see these inside the crystal in the photo. Seeing sets of parallel cracks inside the crystal is very useful for telling you that you are looking at cleavage.





Examples: 5. Quartz, SiO_2

- **Quartz** has no cleavage. It has a "framework" structure and there are no planes across which the bonding is weak. The bonding is mainly covalent, which is inherently strong.
- Quartz fractures conchoidally, like glass. ("conchoidally" = with curved surfaces and concentric lines like growth rings on a shell)
- The absence of cleavage means you will never see fine parallel cracks in quartz crystals, such as you would expect to see in, e.g., barite and calcite.

